

Scalable Data Engineering Solution

Data Engineering I (1TD169) – Spring 2026

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1 Background

1.1 Application Domain and Scientific Context

Urban transportation systems are a primary pillar of modern smart cities, generating vast volumes of heterogeneous data that can be leveraged to improve traffic management, infrastructure planning, and mobility services. The domain of urban mobility is increasingly dependent on large-scale data to solve logistical challenges in densely populated areas. One of the most widely used and comprehensive real-world datasets in this domain is the New York City Taxi Trip dataset, published by the New York City Taxi and Limousine Commission (TLC) [1]. This dataset provides a granular view of city-wide movement by recording millions of individual trips, including precise pickup and drop-off timestamps, geographic zones, trip distances, and detailed fare amounts.

1.2 Dataset Importance and Research Relevance

The NYC taxi dataset is particularly important for data engineering research because it serves as a high-velocity and high-volume benchmark that mirrors the complexity of real-world "Big Data." Its importance lies not only in its scale but also in its ability to reflect human behavior and urban dynamics. Prior research has demonstrated the utility of taxi trip data in various high-impact applications, such as demand prediction, traffic flow analysis, and anomaly detection. For instance, studies in urban computing have utilized mobility datasets to identify spatial and temporal demand patterns, enabling more efficient resource allocation and reduced carbon emissions in cities [2]. Furthermore, such datasets have been used to explore predictive modeling techniques for trip durations and to detect irregular travel patterns.

1.3 Analysis Objective and Engineering Focus

In this project, the analytical objective is intentionally simple: to compute the average trip duration and average fare amount, grouped by pickup zone and hour of the day. This type of aggregation represents a standard workload in data engineering pipelines, involving essential operations such as filtering, grouping, and joining large datasets. By maintaining a straightforward analysis goal, we ensure that the focus remains on the critical engineering aspects—specifically system architecture, performance optimization, and scalability—rather than algorithmic complexity. This approach aligns with the core principles of the course, which emphasize the "heavy lifting" of data processing over advanced statistical modeling.

1.4 Distributed Processing and Scalability Framework

To process this dataset efficiently, the project utilizes a modern big data stack consisting of Apache Spark for distributed computation [3] and the Hadoop Distributed File System (HDFS) for scalable storage. These technologies allow for parallel execution across multiple nodes, which is a necessity given that large-scale data challenges have historically been addressed using paradigms like MapReduce [4]. The system is deployed on a constrained cluster environment consisting of virtual machines, reflecting realistic limitations in computational resources. A central focus of this project is the evaluation of system scalability through both horizontal and vertical scaling experiments. The observed performance results are interpreted using theoretical models such as Amdahl's Law [5], providing a foundation for understanding the limits of parallel speedup in a distributed system.

2 Data Format & Characteristics

2.1 Storage Format and Justification

The dataset utilized in this project consists of New York City Taxi Trip records stored natively in the Apache Parquet format. Parquet, a columnar storage format, was deliberately chosen over row-based alternatives such as CSV because it enables significant data compression and minimizes I/O overhead during analytical query execution. This efficiency is critical for large-scale data processing and essential for operating within the strict 100 GB total storage quota imposed on the OpenStack cluster.

2.2 Dataset Scale and Volume

In total, the raw dataset comprises 144 individual Parquet files spanning from early 2014 to late 2025. After rigorous cleaning and standardization, the final operational dataset yields exactly 219,107,040 valid trip records. For the scope of this project, a subset of approximately 10–20 GB of data is processed. These characteristics are summarized in Table 1.

Table 1: Dataset characteristics.

Property	Value
Format	Apache Parquet
Total Subset Size	10–20 GB
Number of Records	219,107,040
Initial Columns	20
Processed Columns	4

2.3 Schema Structure and Resource Constraints

While the raw TLC schema contains up to 20 diverse attributes, our ETL pipeline aggressively prunes the structure to four essential columns to conserve memory and bypass the 25 GB per-VM storage limitation. The final refined schema is narrowed down to:

- `pickup_datetime`: The date and time when the meter was engaged.
- `PULocationID`: TLC Taxi Zone in which the taximeter was engaged.
- `fare_amount`: The time-and-distance fare calculated by the meter.
- `pickup_hour`: A derived attribute representing the hour of the day for aggregation.

2.4 Data Quality Challenges and Remediation

Several prominent data quality issues were encountered during the ingestion phase. Most notably, schema volatility across different months caused PySpark’s vectorized reader to crash with a schema merge exception. This anomaly was driven by the `Airport_fee` column dynamically shifting its data type between `INT32` and `FLOAT64` depending on the month.

To circumvent this, the pipeline was re-engineered to read files individually, bypassing global schema inference, and explicitly casting core columns to strict unified types. Furthermore, the raw data contained logically invalid operational records; consequently, strict filtering was applied to systematically eliminate trips with zero or negative fare amounts, missing location IDs, and zero-passenger counts prior to aggregation.

3 System Architecture & Implementation

This section details the system architecture, infrastructure deployment, and technical implementation of our robust data pipeline. We outline the hardware topology deployed on OpenStack, why we selected our core software stack, and the engineering mechanisms driving our ETL and analysis jobs.

3.1 System Design

Our system follows a classic centralized-master, distributed-worker model tailored for batch processing big data. As depicted in Figure 1, the architecture is divided into three primary layers: Ingestion, Distributed Storage (HDFS), and Distributed Processing (Apache Spark). Data is fetched externally into the master node, pushed to HDFS, and subsequently processed by Spark executors dispersed across the worker nodes.

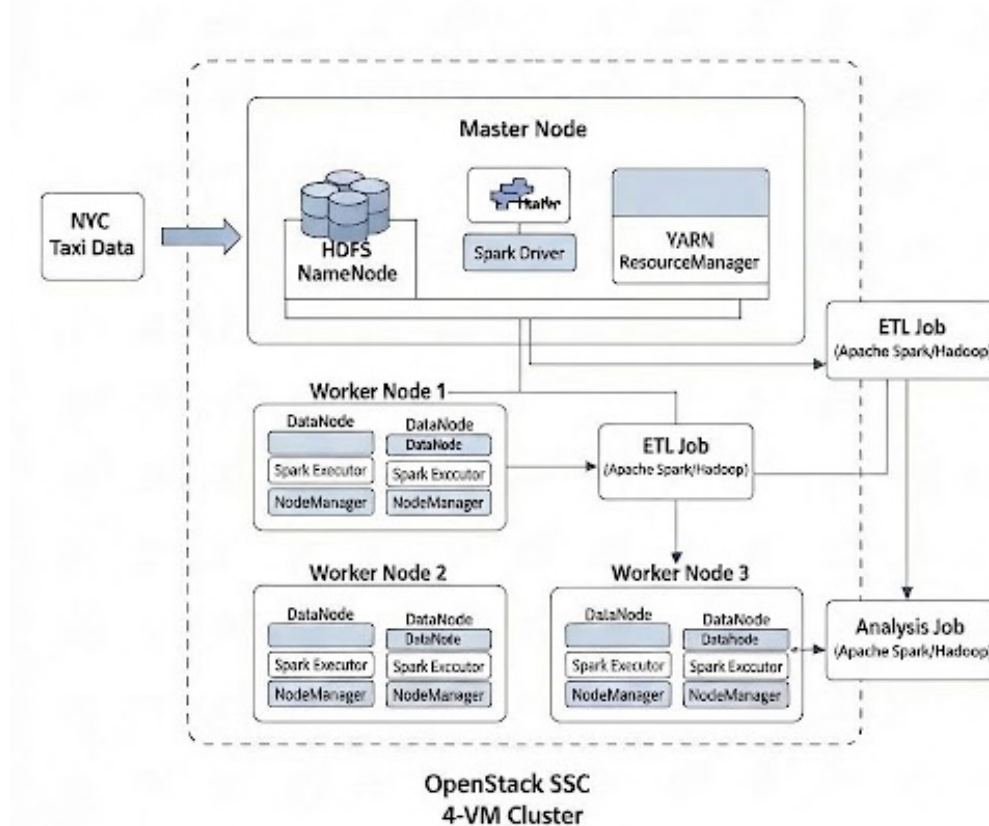


Figure 1: System architecture diagram showing the OpenStack deployment, node topology, and the internal data pipeline flow from raw Parquet to aggregated results.

3.2 Technology Stack

Given the project requirements, we adopted a modern big data stack focused on in-memory processing capabilities:

- **Storage (HDFS via Hadoop 3.4.1):** Chosen for its fault tolerance and native integration with Spark. While cloud object storage (like AWS S3) is popular today, deploying HDFS on bare VMs offers direct control over data locality and block replication factors.
- **Processing (Apache Spark & PySpark 3.5.1):** We selected Spark over standard Hadoop MapReduce because Spark holds intermediate RDDs and DataFrames in mem-

ory, drastically accelerating the iterative transformations required during our ETL phase. PySpark provides an accessible DataFrame API and straightforward type-casting structures.

- **Orchestration & Resource Management:** We relied on standalone cluster management managed heavily by Bash scripts (`run_benchmark.sh` and `deploy.sh`) combined with Git for CI/CD, creating a lightweight yet effective pipeline orchestration.

3.3 Infrastructure Configuration

Our cluster is deployed entirely on the Swedish National Infrastructure for Computing (SNIC) OpenStack instance. The infrastructure consists of 4 tightly coupled Virtual Machines (VMs):

- **Master Node (1x ssc.medium):** 2 vCPU, 4 GB RAM. This node is assigned a Floating IP for external SSH access. It houses the Hadoop NameNode, ResourceManager, and the Spark Driver process.
- **Worker Nodes (3x ssc.medium):** 2 vCPU, 4 GB RAM each. These sit on a private sub-network (192.168.2.x subnet). They run the Hadoop DataNodes, NodeManagers, and Spark Executor instances.

In total, the cluster harnesses 8 vCPUs and 16 GB of RAM, capped with a strict 100 GB collective storage limit. To maintain data reliability without exceeding storage quotas, HDFS was configured with a replica factor of 2 (`dfs.replication=2`).

3.4 Data Pipeline

The pipeline executes in three synchronized stages:

1. **Ingestion:** The data ingestion and core ETL pipeline were engineered to handle large-scale NYC Taxi data efficiently. An automated Bash script (`scripts/ingest_NYC_taxi.sh`) was developed to manage the creation of staging directories, concurrently download 12 months of Parquet files, and verify file integrity by removing corrupted downloads smaller than 100 KB. This script streamlined the ingestion process by depositing raw data directly into the Hadoop Distributed File System (HDFS) while maintaining strict directory structures and adhering to cluster storage constraints.
2. **ETL Phase:** The `etl_job.py` script ingests the raw data block by block. It filters out invalid records (e.g., zero-distance trips, null metrics), converts timestamps to calculate `duration_minutes` and `pickup_hour`, aggressively prunes dropped columns, and commits the clean records back to HDFS at `/data/nyc-taxi-cleaned/`.
3. **Analysis Phase:** Finally, `analysis_job.py` aggregates the validated dataset. It groups by `PULocationID` and `pickup_hour` to compute the average fare and trip duration. It subsequently performs a left join against a static CSV (`taxi_zone_lookup.csv`) to convert abstract ID numbers into human-readable zone names, coalescing the final output into a single file for stakeholder ingestion.

3.5 Code Structure

The codebase follows a modular design paradigm:

- **src/config.py:** Centralizes cluster endpoints, absolute HDFS URI paths, and basic threshold configurations to eliminate hardcoded parameters across jobs.

- **scripts/ingest_nyc_taxi.sh**: Automates the initial data acquisition phase. It manages staging directories, concurrently downloads the TLC Parquet files, verifies file integrity by purging corrupted downloads, and directly deposits the raw dataset into HDFS.
- **notebooks/final_notebook.ipynb**: Serves as the primary development and validation environment for the core ETL logic. It was used to prototype strict schema enforcement, data filtering, and geographic `.join()` operations prior to modularization into the production `src` scripts.
- **src/etl_job.py**: The heaviest processor. Given resource constraints, it bypasses standard `spark.read.parquet("*.parquet")` schema inferences and iteratively targets singular files using `subprocess.run` to interact with HDFS.
- **src/analysis_job.py**: Houses the PySpark SQL aggregations, strictly outputting to a targeted CSV path using `coalesce(1)`.
- **scripts/run_benchmark.sh**: A specialized harness that automates execution over varied core allocations (`-total-executor-cores`) and node arrays to capture timestamps for plotting horizontal and vertical scales.

3.6 Challenges & Solutions

We encountered several prominent technical challenges during development:

1. **Data Ingestion Access Blocks**: During the data ingestion phase, automated acquisition of the multi-year NYC Taxi Parquet datasets was frequently interrupted by AWS CloudFront 403 (Forbidden) errors. These blocks prevented the script from establishing a stable connection to the TLC data repository. *Solution*: To resolve this, a robust HTTP request strategy was implemented. By configuring a specific **User-Agent** header within the ingestion scripts, the system successfully bypassed the CloudFront security filters, ensuring consistent and uninterrupted dataset downloads.
2. **Dataset Schema Volatility**: When reading multi-month TLC data, PySpark's vectorized Parquet reader routinely crashed (`[CANNOT_MERGE_SCHEMAS]`) due to invisible shifts in the column type for `Airport_fee` (varying from `INT32` to `FLOAT64` dynamically by month). *Solution*: We restructured the ETL job into a `for`-loop that asks HDFS for the explicit list of file paths. Spark reads them independently, safely infers local schemas, selectively prunes the offensive column, and appends into a normalized structure.
3. **Strict Storage Exhaustion**: 25 GB of storage per VM meant HDFS was easily overwhelmed caching uncompressed RDDs and saving multiple layers of Parquet processing data. *Solution*: We established an aggressive "select and drop" mechanism in the ETL. We discard roughly 15 unused columns representing several gigabytes of overhead, saving only the 4 tightly packed elements (`PULocationID`, `fare_amount`, `pickup_hour`, `duration_minutes`) needed for analysis.
4. **Dynamic Allocation Skewing Benchmark Constraints**: Spark's dynamic resource allocation skewed the early iterations of our benchmarking by automatically distributing workloads across nodes we were attempting to restrict. *Solution*: We disabled dynamic allocation directly via PySpark config keys (`spark.dynamicAllocation.enabled=false`) and enforced hard caps using Spark submit metrics.

4 Scalability Experiments & Results

[Present experimental findings: experimental design, results with plots, analysis of results, bottleneck identification, and comparison to theoretical expectations.]

4.1 Experimental Design

[Describe what you varied in experiments (number of workers, cores, data size), the metrics measured (runtime, throughput), and the methodology used.]

4.2 Horizontal Scaling Results

[Present results of horizontal (strong) scaling experiments: fixed dataset size with increasing number of worker nodes. Include runtime vs. number of workers plot.]

4.3 Vertical Scaling Results

[Present results of vertical scaling experiments: fixed number of nodes with increasing cores/memory per node. Include runtime vs. resources plot.]

4.4 Speedup & Efficiency Analysis

[Compute and plot the speedup factor ($S = T_1/T_N$) and efficiency ($E = S/N$). Compare against ideal linear speedup. Discuss deviations.]

4.5 Bottleneck Identification

[Identify what limits scalability: CPU, network I/O, disk I/O, shuffle overhead, data skew, or serialization. Reference Amdahl's Law if applicable.]

5 Discussion & Conclusion

[Reflect on the project: Was the architecture suitable? What worked well? What would you improve with more time/resources? What did you learn about distributed systems?]

References

- [1] New York City Taxi and Limousine Commission, *New York City Taxi and Limousine Commission (TLC) Trip Record Data*, Accessed: 2026-03-01, 2024. [Online]. Available: <https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>
- [2] G. Andrienko, N. Andrienko, P. Bak, D. Keim, and S. Wrobel, “Visual analytics of movement,” 2013.
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- [4] J. Dean and S. Ghemawat, “Mapreduce: Simplified data processing on large clusters,” *Communications of the ACM*, vol. 51, no. 1, pp. 107–113, 2008.
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